Unit Test Report

Project: Online Quiz System
Environment: MSYS2 UCRT64, GCC 16.2.0
Test result: 9/9 test programs passed

Executed Test Areas

Area	Coverage
Quiz catalogue and question parsing	Valid, malformed, missing, and invalid records
Score calculation	Correct, partial, zero, lowercase, empty, and mismatched answers
Administrator management	Add, edit, delete, duplicate, invalid, and rollback paths
Registration and authentication	Valid, duplicate, invalid, malformed, and missing-file paths
Security logging	Normal events, sanitization, truncation, and write failures
Interactive application	Registration, login, quiz, logout, cancellation, and menu errors
TCP submission client	Invalid input, connection errors, and OK/ERROR responses
Quiz results server	Parsing, persistence, loopback handling, and oversized requests

Conclusion

All nine test programs completed successfully in the MSYS2 UCRT64 environment. No test assertion failures were reported.